Solving multi-step equations

1 Consider the equation $9\,(n+5) = 4n + 50$ for $n = 1$.

- **a** Find the value of the left-hand side of the equation when $n = 1$.
- **b** Find the value of the right-hand side of the equation when $n = 1$.
- **c** Consider your answers from parts (a) and (b).
  Is $n = 1$ a solution of $9\,(n+5) = 4n + 50$?

2 Consider the equation $0.28n - 0.4 = 2.3n - 3.03$ for $n = 0.8$.

- **a** Find the value of the left-hand side of the equation when $n = 0.8$.
- **b** Find the value of the right-hand side of the equation when $n = 0.8$.
- **c** Consider your answers from parts (a) and (b).
  Is $n = 0.8$ a solution of the equation $0.28n - 0.4 = 2.3n - 3.03$?

3 Determine whether the given value of the variable is a solution to the equation.

- **a** $p = 0;\ 3p - 9 = 7p - 1$
- **b** $x = -2;\ 3x - (4x - 8) = -6$
- **c** $x = \dfrac{7}{12};\ x - \dfrac{4}{9} = \dfrac{17}{36}$
- **d** $x = -3;\ \dfrac{-24}{x} + 7 = 5x$
- **e** $x = \dfrac{1}{4};\ x + \dfrac{2}{3} = \dfrac{11}{12}$

4 Complete the following equation such that $x = -7$ is the only solution.
$2\,(x - 12) + 9 = 3x - \square$

5 Complete the following equation such that it has no solutions.
$80x + \dfrac{1}{6} = \square\,(8x + 2)$

6 Solve the following equations:

- **a** $8k + 2 = 15k - 12$
- **b** $8r - 2 = 4r + 14$
- **c** $10r + 8 = 74 - r$
- **d** $12r + 8 = 116 - 6r$
- **e** $36 - 2k = 6k + 20$
- **f** $-90 - 8v = -17v + 9$
- **g** $\dfrac{14k - 76}{2} - 2 = 2k$
- **h** $-n - \dfrac{1}{3} = 2$
- **i** $\dfrac{y - 4.3}{-4} = 7.7$

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7 Solve for x:

a $6x + 9 = 10x - 3$

b $-\frac{6x}{5} + 5 = 35$

c $\frac{11x + 22}{8} = 33$

d $3x - 8 = x$

e $5x + 2 = 3x + 22$

f $-107 - 8x = -17x + 19$

g $\frac{5x}{6} + 4 = 21.5$

h $6x - \frac{11}{7} = \frac{409}{7}$

i $\frac{11.8x + 31.86}{4} = -106.2$

j $\frac{4x - 20}{7} = \frac{28}{3}$

k $x - \frac{1}{5} = \frac{3}{4}$

l $7x - \frac{90}{7} = x$

m $8x - \frac{112}{3} = 4x$

n $x - \frac{5}{12} = -\frac{1}{4}$

o $12x + 19 = x + 85$

p $9x + 4 = 10x + 2$

8 Find the value of x.

a $3(x + 1) + 2 = -1$

b $-2(x + 2) - 4 = 4$

c $3(x + 6) + 3(x + 24) = 12$

d $2(2x + 5) = 3(x + 5)$

e $-7(x + 1) = -28$

f $3(x - 1) - 5 = -8.6$

g $3.8(x + 4) - 2 = 1.8$

h $3\left(x + \frac{3}{2}\right) + 3 = -\frac{15}{2}$

i $5.3(3x + 55) = 4.9(x + 55)$

j $4(5x + 1) = -3(5x - 5)$

k $5(2x + 2) = -3(4x - 5) + 5x$

9 Determine the number of solution each of the following equations has.

a $18 + 3x = 3x - 16$

b $8x - 4x = 4$

c $x + 8 = x + 6$

d $2x - 2 = 2x - 2$

e $7x + 4 = 6x + 5$

f $8x = 8x + 8$

g $6x + 5 = 7x - 5$

h $9x - 9 = 9x + 7$

i $7x + 7 = 9x - 7$

j $2x = 2x + 4$

k $8x + 6 = 8x + 6$

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l $10\left(5+x\right)=10\left(x+5\right)$

m $\dfrac{10+x}{10}=\dfrac{x+4}{10}$

n $7\left(7+x\right)=2x+28$

o $-31-3\left(x-9\right)=2\left(x-2\right)-5x$

p $24+7\left(x-23\right)+3=16+8\left(x+14\right)-x-262$

10 Consider the equation $3x-29=9x+13$.

a Solve for x.

b How many solutions does the equation $3x-29=9x+13$ have?

11 For each of the following relations:

i Using n to represent the unknown number, form an equation that represents this relation.

ii Determine the number of solutions the relationship has.

a Three more than three times a number is equal to six more than four times the number

b Six more than two times a number is equal to five more than two times the number

c Two more than four times a number is equal to two added to quadruple the number

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